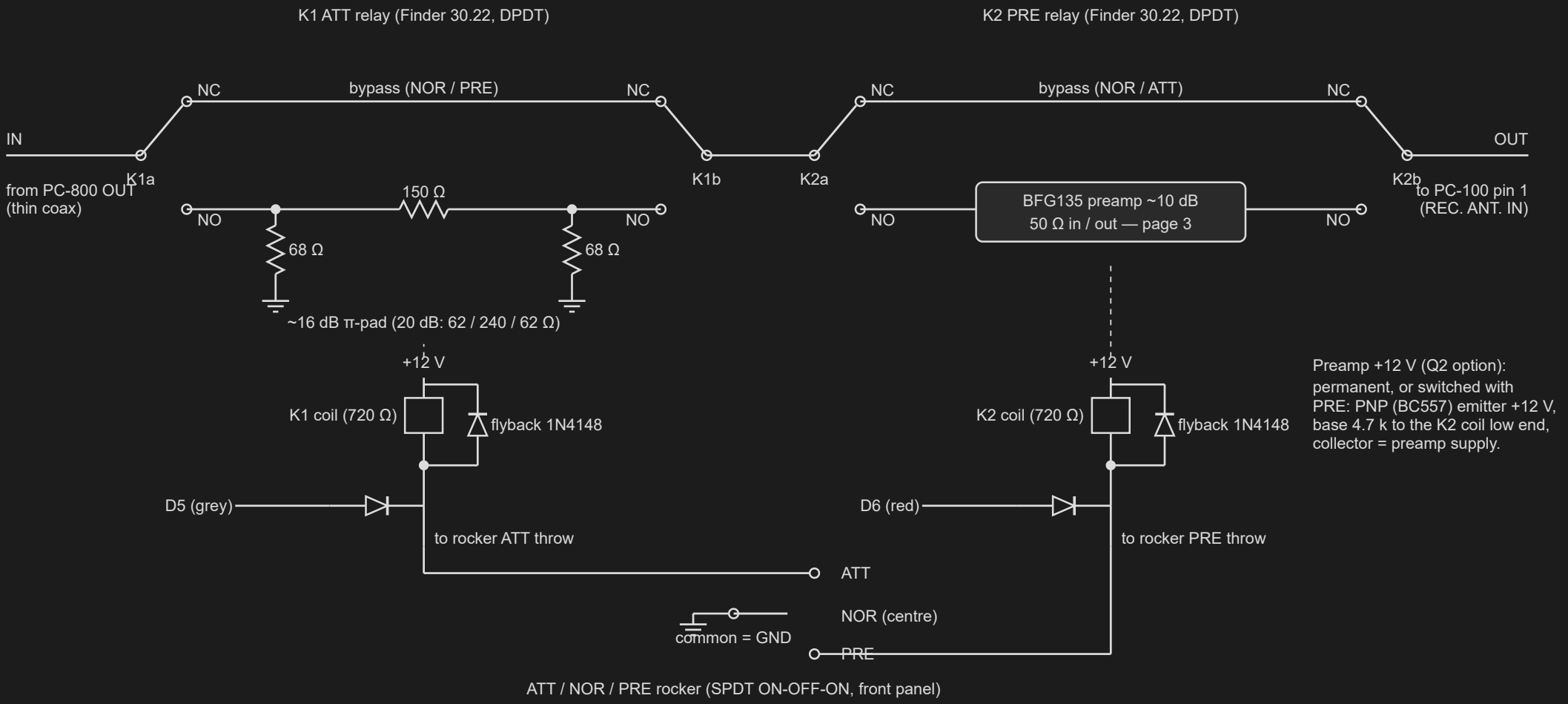
ATT / PRE board — relays, ~16 dB attenuator, BFG135 preamp

YT3CT · 2026-10 · proposal · RX line between PC-800 OUT and PC-100 pin 1 (after the preselector). RX only — never in the TX path.



- Relays at rest (both coils off) = **NOR**: signal goes IN → K1 bypass → K2 bypass → OUT.
- ATT: K1 pulls, both poles move to NO → signal through the π-pad. PRE: K2 pulls → signal through the preamp. The rocker can only select one.
- Coil high ends permanently +12 V; the rocker grounds the coil low end. Flyback 1N4148 across each coil (band to +12 V). Sense 1N4148 from the coil low end to D5 / D6 (band to the coil) — Nano reads LOW = on.
- Thin coax (RG-174 / RG-316) in and out, short connections on the board, ground plane; keep the bypass track short and away from the preamp output.
- Check before cutting: the PC-800 → PC-100 coax must carry receive only (no TX signal).

ATT / PRE board — PCB design notes

Everything the board needs, for drawing the PCB. Values as on pages 1 and 3.

Connectors (board edge)

Ref	Signal	Goes to
J1	RF IN + shield	coax from PC-800 OUT
J2	RF OUT + shield	coax to PC-100 pin 1
J3-1	+12 V	radio +12 V (coils, preamp)
J3-2	GND	chassis
J3-3	K1 coil low end	rocker ATT throw
J3-4	K2 coil low end	rocker PRE throw
J3-5	ATT sense	Nano D5 (grey)
J3-6	PRE sense	Nano D6 (red)

Operation

Rocker	K1	K2	Signal path	D5 / D6
NOR	off	off	bypass – bypass	HIGH / HIGH
ATT	on	off	π-pad (−16 dB) – bypass	LOW / HIGH
PRE	off	on	bypass – preamp (+10 dB)	HIGH / LOW

Parts list (whole board)

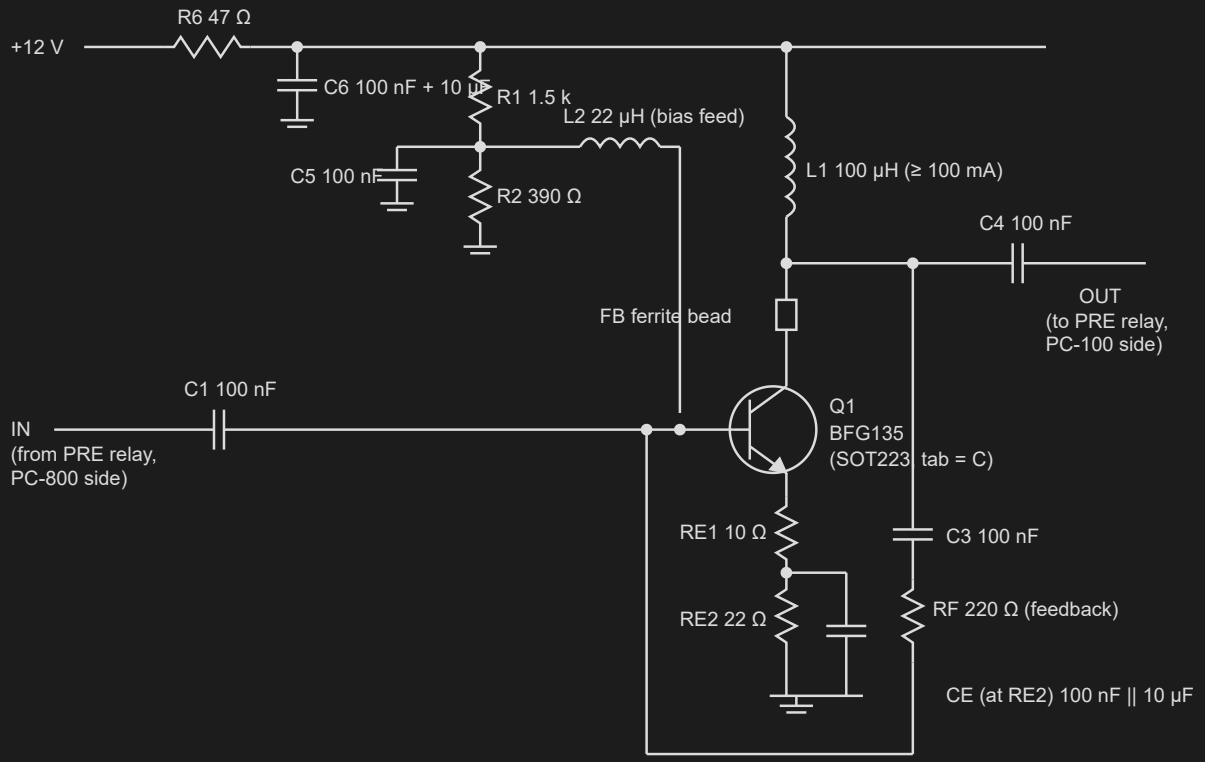
Ref	Part	Note
K1, K2	Finder 30.22.7.012 DPDT 12 V	720 Ω coil, AgNi+Au, 5 mm pitch
D1, D2	1N4148	flyback across K1 / K2 (band to +12 V)
D3, D4	1N4148	sense: anode to J3-5 / J3-6, band to the coil low end
R10, R12	68 Ω	π-pad shunt (20 dB version: 62 Ω)
R11	150 Ω	π-pad series (20 dB version: 240 Ω) — 15.9 dB, Z_{in} 49.3 Ω
Q1 + preamp	BFG135 + parts of page 3	R1 1.5 k, R2 390, RF 220, RE1 10, RE2 22, R6 47, L1 100 µH, L2 22 µH, C1–C6, CE, ferrite bead
Q2 (option)	BC557 + R13 4.7 k	switches the preamp supply with PRE
C7	100 nF + 10 µF	+12 V decoupling at J3

Layout

- Double-sided, bottom = solid ground plane with many vias. RF tracks short and wide.
- Signal flow in a line: J1 → K1 → K2 → J2; pad between the K1 poles, preamp between the K2 poles.
- Preamp input and output apart (no loop); BFG135 tab on a copper area; ferrite bead right at the collector.
- Coils, sense diodes and supply in a corner near J3, away from the RF path.
- Coax shields to the ground plane right at J1 / J2.

RX preamp — BFG135 broadband feedback amplifier

YT3CT · 2026-10 · proposal · ~10–11 dB, 50 Ω in / out, 3.5–30 MHz, I_C ≈ 50 mA at 12 V · sits on the PRE relay between PC-800 OUT and PC-100 pin 1



Parts

Part	Value	Note
Q1	BFG135	SOT223, collector tab soldered to a copper pad (heat, ~0.4 W)
RF	220 Ω	sets gain / 50 Ω match (RF = Z0 · (1 + gain))
RE1	10 Ω	unbypassed, RF feedback (RE1 = $Z0^2$ / RF)
RE2 / CE	22 Ω / 100 nF ‖ 10 µF	DC stabilisation, bypassed for RF
R1 / R2	1.5 k / 390 Ω	base ≈ 2.4 V → I_C ≈ 50 mA
L1	100 µH	collector choke, rated ≥ 100 mA
L2	22 µH	keeps the divider off the input
R6 / C6	47 Ω / 100 nF + 10 µF	supply filter
C1, C3, C4, C5	100 nF	ceramic, short leads
FB	ferrite bead	on the collector lead — against VHF oscillation

Notes

- Expected: **~10–11 dB** gain, 50 Ω in / out, flat 3.5–30 MHz. More gain: RF 270 Ω + RE1 8.2 Ω (~12 dB).
- Check after building: collector ≈ 9.6 V, emitter ≈ 1.6 V (= 50 mA). Adjust R2 if far off.
- BFG135 has f_T ≈ 7 GHz: build on a ground-plane board, very short leads, all ground returns to the plane.
- Supply: permanent +12 V, or switched on with PRE (PNP from the PRE coil low end) to save ~60 mA.
- On the PRE relay: pole 1 switches the input, pole 2 the output (bypass when off). RX only — never in the TX path.
- Test with the tinySA (sweep through the amp, 50 Ω both ends) or by ear on 10 / 15 m.